BCS-080	Digital Image Processing	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	To learn the advanced concepts of image processing and its implementation.
CO2	Enhance the visual quality of given grey/color image using well known transformations and filters.
CO3	Distinguish between lossy and lossless image compression prototypes.
CO4	Demonstrate the use of MATLAB to create correlative image processing applications
CO5	To learn about Max and Min filters

Detailed Syllabus

MODULE I

Introduction and Fundamentals Motivation and Perspective, Applications, Components of Image Processing System, Element of Visual Perception, A Simple Image Model, Sampling and Quantization. Image Enhancement in Frequency Domain Fourier Transform and the Frequency Domain, Basis of Filtering in Frequency Domain, Filters – Low-pass, High-pass; Correspondence Between Filtering in Spatial and Frequency Domain; Smoothing Frequency Domain Filters – Gaussian Lowpass Filters; Sharpening FrequencyDomain Filters – Gaussian Highpass Filters; Homomorphic Filtering.

MODULE II

Image Enhancement in Spatial Domain Introduction; Basic Gray Level Functions – Piecewise-Linear Transformation Functions: Contrast Stretching; Histogram Specification; Histogram Equalization; Local Enhancement; Enhancement using Arithmetic/Logic Operations – Image Subtraction, Image Averaging; Basics of Spatial Filtering; Smoothing - Mean filter, Ordered Statistic Filter; Sharpening – The Laplacian.

MODULE III

Image Restoration A Model of Restoration Process, Noise Models, Restoration in the presence of Noise only-Spatial Filtering – Mean Filters: Arithmetic Mean filter, Geometric Mean Filter, Order Statistic Filters – Median Filter, Max and Min filters; Periodic Noise Reduction by Frequency Domain Filtering – Bandpass Filters; Minimum Mean-square Error Restoration.

Course Outcomes: After the completion of the course the student will be able to:

CO1	To acquire the knowledge of soft computing and hard computing
CO2	To develop skill in soft computing methodology
CO3	To acquire the knowledge of the fuzzy Neural network and Genetic Language
CO4	To analyze and optimized the problem of real-life applications
CO5	Able to explain Max and Min filters

Text Books:

1. Digital Image Processing 2nd Edition, Rafael C. Gonzalvez and Richard E. Woods. Published by: Pearson Education.
2. Digital Image Processing and Computer Vision, R.J. Schalkoff. Published by: John Wiley and Sons, NY.
3. Fundamentals of Digital Image Processing, A.K. Jain. Published by Prentice Hall, Upper Saddle River, NJ.
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